

TERMINALFIX DUI70.DLL SIDELOAD HUNT

Campaign-specific hunt for a legitimate Lock Screen binary loading a co-located malicious DirectUI DLL outside its standard Windows SystemApps location.

HUNTING QUERY

CONFIDENCE: MEDIUM

LockScreenContentServer.exe

legitimate signed sideload host



dui70.dll

same directory as host

Outside standard location

C:\Windows\SystemApps\

REPORT DATE

30 August 2026

STATUS

Untested implementation sketch

PRIMARY TELEMETRY

Defender XDR /
DeviceImageLoadEvents

ATT&CK

T1574.002 - DLL Side-
Loading

A match identifies the source-observed process-module-location pattern. It does not prove the complete TerminalFix chain without file and incident validation.



EXECUTIVE ASSESSMENT

Decision-grade summary and analytical boundaries.

ASSESSMENT

OPERATIONAL VALUE

9.4

Direct module-load telemetry for a high-specificity sideload pair in an active multistage intrusion.

CLASSIFICATION

Campaign-specific artifact

Hunting query - Untested implementation sketch. Promotion requires local prevalence and ownership validation.

1published query
0production rules
Mediumconfidence
MDEprimary platform

Source fact

Microsoft observed LockScreenContentServer.exe load a malicious co-located dui70.dll after a TerminalFix PowerShell stage.

Analytical meaning

The legitimate host, exact imported DLL, co-location and non-standard path form a strong prevalence pivot with low expected routine volume.

Assumptions

Defender captures the module load and the legitimate enterprise baseline keeps this binary under SystemApps.

Promotion boundary

Do not alert or isolate automatically until legitimate copied binaries, servicing behavior and test tooling are measured.



THREAT MECHANICS

Source-supported chain and the exact observable boundary.

BEHAVIOR

01 TerminalFix lure

Fake CAPTCHA directs the user to paste PowerShell.
→

02 Archive staging

A ZIP is extracted under C:\ProgramData.
→

03 Host launch

A batch file starts the signed Lock Screen binary.

04 DLL sideload

The host resolves co-located dui70.dll.
→

05 Second stage

Obfuscated code retrieves steganographic payloads.
→

06 Reverse tunnel

A Python implant enables internal TCP pivoting.

Condition	Evidence	Query use
LockScreenContentServer.exe	Observed sideload host	Required
dui70.dll	Observed malicious DLL name	Required
Same directory	Observed loader resolution	Required
Outside SystemApps	Microsoft guidance	Required

ATT&CK

T1574.002 - DLL Side-Loading. The hunt observes the sideload condition, not the downstream reconnaissance or tunnel.



TELEMETRY & EVIDENCE MODEL

Minimum data contract and platform limits.

TELEMETRY

REQUIRED

DeviceImageLoadEvents

Microsoft Defender for Endpoint DLL load telemetry from onboarded Windows endpoints.

OPTIONAL ENRICHMENT

SHA1, initiating command line, process identity, account, signer, file creation, network and adjacent PowerShell telemetry.

Field	Reliability	Purpose
Timestamp, DeviceId, DeviceName	Documented	Event and device pivot
FileName, FolderPath	Documented	Loaded module identity and path
InitiatingProcessFileName	Documented	Sideload host identity
InitiatingProcessFolderPath	Documented	Co-location and standard-path check
InitiatingProcessUniquelId	Documented	Strong process pivot
SHA1, ReportId	Documented	File and event enrichment

Unavailable through this hunt

DLL content, successful payload execution, steganographic extraction, persistence, reconnaissance, tunnel establishment and attribution.

Coverage gap

Disabled image-load telemetry, an unonboarded host, a renamed host/DLL pair or a different sideload dependency produces a false negative.



DETECTION LOGIC

Direct module-load filters with no threshold, join or rarity assumption.

KQL

HYPOTHESIS

If TerminalFix executes the observed DLL sideload, module-load telemetry should show LockScreenContentServer.exe loading a same-directory dui70.dll outside SystemApps. Legitimate overlap is resolved through path ownership, signer, hash and deployment context.

KQL - UNTESTED HUNTING

```
DeviceImageLoadEvents
| where InitiatingProcessFileName =~ "LockScreenContentServer.exe"
| where FileName =~ "dui70.dll"
| extend LoadedDllDirectory = tostring(parse_path(FolderPath).DirectoryPath)
| where LoadedDllDirectory =~ InitiatingProcessFolderPath
| where not(InitiatingProcessFolderPath startswith @"C:\Windows\SystemApps\")
| project Timestamp, DeviceId, DeviceName,
    InitiatingProcessFileName, InitiatingProcessFolderPath,
    InitiatingProcessCommandLine, InitiatingProcessUniqueId,
    FileName, FolderPath, SHA1,
    InitiatingProcessAccountName, InitiatingProcessAccountSid,
    ReportId
| order by Timestamp desc
```

THRESHOLD	WINDOW	JOIN	KEY	STATUS
None	Retention	None	ProcessUniqueId	Untested

Why Defender XDR

Direct DLL-load telemetry, documented path fields and a strong device/process pivot. No Sentinel translation or third-party parser is assumed.



TRIAGE & INVESTIGATION

Convert a match into defensible incident evidence.

WORKFLOW

1

Confirm load

Verify host path, DLL path, same-directory equality, time and process identity.

2

Validate files

Collect hashes, signatures, version metadata and safe binary copies.

3

Pivot backward

Review archive extraction, batch launch and preceding PowerShell activity.

4

Pivot forward

Hunt PNG retrieval, Run keys, scheduled tasks, AD discovery and Python execution.

5

Scope network

Treat confirmed hosts as potential pivots and inspect credentials and lateral access.

Escalate

Unsigned or source-matching DLL, ProgramData staging, PowerShell ancestry or tunnel evidence.

Benign

Approved software packaging or servicing with verified Microsoft files and change record.

Inconclusive

Missing binaries, expired telemetry or incomplete signer/hash enrichment.

Response boundary

A corroborated match warrants isolation and credential review because the source-observed chain creates a network pivot. Filename-only evidence does not.



FALSE POSITIVES & VALIDATION

What must be proven before alerting.

QUALITY CONTROL

SCENARIO	LIKELIHOOD	DISPOSITION
Application packaging or compatibility test	M	Verify owner, signer and change record
Windows servicing residue	L	Validate trusted Microsoft files and path
Malware-analysis lab execution	L	Allowlist isolated lab devices only
TerminalFix sideload	L	Emergency investigation

Validation steps

1. Run the hunt across full available retention.
2. Inventory all non-SystemApps executions of the host.
3. Generate a safe lab image-load sample with copied benign files.
4. Confirm co-location logic, field population and investigator pivots.

Decision criteria

Success: lab load is visible and production matches are low-volume and dispositionable.

Rejection: common managed software creates the same compound pattern.

Failure: image-load events or path fields are absent.

Allowlisting rule

Allowlist only verified file hashes/signers plus stable deployment paths and owners. Never suppress every non-standard LockScreenContentServer execution.



CONCLUSION & REFERENCES

Coverage statement, platform limits and document control.

FINAL

CONCLUSION

A strong campaign hunt with a clear promotion experiment.

The exact process-module co-location and non-standard path are directly observed and supported by documented Defender telemetry. Production promotion remains contingent on local benign prevalence and safe validation.

Cross-SIEM decision

Splunk: no reliable implementation without a verified EDR raw source, index, sourcetype and module-load field extraction.

Chronicle: no reliable YARA-L rule without a verified parser and UDM PROCESS_MODULE_LOAD mapping for host and module paths.

Deferred ideas

Exact C2 domains, Python client arguments, random persistence names, generic ClickFix PowerShell and PNG traffic. Each is IOC-dependent, incomplete, or has insufficient local mapping for this dossier.

References

1. Microsoft Security Research, 28 Aug 2026 - TerminalFix campaign deploys a reverse tunnel through multistage intrusion.
[microsoft\[.\]com/en-us/security/blog/2026/08/28/terminalfix-campaign-deploys-reverse-tunnel-through-multistage-intrusion/](https://microsoft.com/en-us/security/blog/2026/08/28/terminalfix-campaign-deploys-reverse-tunnel-through-multistage-intrusion/)
2. Microsoft Learn - DeviceImageLoadEvents, updated 11 Jun 2025.
[learn\[.\]microsoft\[.\]com/en-us/defender-xdr/advanced-hunting-deviceimageloadevents-table](https://learn.microsoft.com/en-us/defender-xdr/advanced-hunting-deviceimageloadevents-table)
3. Microsoft Learn - parse_path(), updated 20 Aug 2024.
[learn\[.\]microsoft\[.\]com/en-us/kusto/query/parse-path-function](https://learn.microsoft.com/en-us/kusto/query/parse-path-function)
4. MITRE ATT&CK - T1574.002 DLL Side-Loading.
[attack\[.\]mitre\[.\]org/techniques/T1574/002/](https://attack.mitre.org/techniques/T1574/002/)

Document control

English dossier. No customer data. References defanged. HTML and rendering assets remain private and are excluded from publication.